NATIONAL MATHEMATICS CONTEST
O-LEVEL PAPER 2.

SATURDAY 17TH JUNE, 2017 9:00 AM - 12:15 PM

INSTRUCTIONS TO CANDIDATES AND SUPERVISORS:

(a) *This competition is conducted on the assumption that proper security is maintained. UMS reserves the right, should there be evidence or suspicion of any malpractice, to reject scripts from a particular school/institution.*

(b) *There are* **two** *sections and each section carries* 50 *marks. Attempt all questions in exactly three hours fifteen minutes (*$9:00am$ *to* $12:15pm$*). Marks will be awarded for only answers for which a clear and logical layout of the working is provided.*

(c) *Answer scripts of only qualified students should be returned to the contest coordinator, latest* **Monday 19th June, 2017**.

(d) **ALL** *participants who qualified and sat for Paper* 2 *are cordially invited to the certificate and/or prize giving ceremony on Saturday* 29*th July,* 2017 *at NOON at Makerere University, Kampala.*

SECTION A : 5 marks each

Qn. 1. If the price of sugar increases from 3,000 UGX per kilogram to 5,000 UGX per kilogram. By what percentage will the amount of sugar bought be reduced?

Qn. 2. When 3430 is divided by the odd number n, the quotient is a positive prime number and the remainder is 0. What is n?

Qn. 3. Given that,

$$\frac{1}{x}+\frac{1}{y}=\frac{1}{a},\ x+y=b \ \text{ and } \ x^2+y^2=c^2.$$

Find the possible relationships between a and b if $c=0$.

Qn. 4. What is the ratio of the area of a circle to the area of an inscribed square?

Qn. 5. Mr. Okello can slash a square kilometre compound in 40 minutes, Chris can do the same job in 30 minutes. How many minutes will it take the two to slash the same compound if they both work at the same rate.

Qn. 6. A lady drives at $40km/h$ while going to visit her uncle. On arrival she realises her uncle had gone on a trip. Immediately she drives back to her home at $90km/h$. What is her average speed for the whole trip?

Qn. 7. In a triangle ABC, D lies on AC, and E lies on BC with DE parallel to AB. If $AC=5$, $DC=4$ and $BE=2$ then what is the length of EC?

Qn. 8. A litre bottle of a liquid contains 10% alcohol, and another 3 litre bottle full of liquid contains 30% alcohol. They are mixed together. What percent of the mixture is not alcohol?

Qn. 9. Suppose that a and x are two positive real numbers for which

$$\log_x a + \log_a x = 3.$$

What is the value of

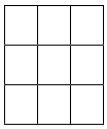
$$(\log_a x)^2 + (\log_x a)^2$$

Qn. 10. Mary paid UGX 480,000 to purchase a certain number of items, but the nice vendor gave her two extra. This decreased the price per item by UGX 1,000. How many items did she receive (including the two extra)?

SECTION B : 10 marks each

Qn. 11. In a group of five friends, the sums of the ages of each group of four of them are 125, 129, 131, 137, and 142. What is the age of the youngest of the friends?

Qn. 12. Suppose that the numbers $1, 2, 4, 8, 16, 32, 64, 128$, and 256 are placed into the 9 squares in a 3×3 grid in such a way that each of the 9 numbers appears exactly once and the product of the numbers appearing in any row or column is the same. What is the value of the product in each row and column?



Qn. 13. In racing over a given distance at uniform speeds, John can beat Isa by 20 meters, Isa can beat David by 10 meters, and John can beat David by 28 meters. How many meters is the distance over which they are racing.

Qn. 14. What is the units digit of 2^{2017}?

Qn. 15. A circle passes through two adjacent vertices of a square and is tangent to the opposite side of the square. If the side length of the square is 2. What is the radius of the circle?

SECTION A : 5 marks each

Qn. 1. Percentage increase in price is

$$\left(\frac{5,000-3,000}{3,000}\right)\times 100 = 66.67\%$$

The new price is 1.6667 times the original price $\Rightarrow$ the amount of sugar that will be bought is

$$\frac{1}{1.6667} = 0.59998 \approx 0.6$$

of a kilogram thus the amount of sugar bought will be reduced by 0.4 that is 40%

Qn. 2.

$$3430 = 343\times 10 = 7^3\times 2\times 5$$

The only way 3430 can be written so as to have an odd number and a prime number is

$$3430 = 2\times(7^3\times 5)$$

this implies then $n = 7^3\times 5 = 1715$.

Qn. 3.

$$\frac{1}{x}+\frac{1}{y}=\frac{1}{a}\Leftrightarrow\frac{x+y}{xy}=\frac{1}{a}$$

since $x+y=b\Rightarrow ab=xy$ then considering

$$(x+y)^2 = x^2+2xy+y^2\Rightarrow b^2=c^2+2ab$$

then with $c=0$ implies

$$b^2=2ab\Rightarrow b(b-2a)=0\Rightarrow b=0$$

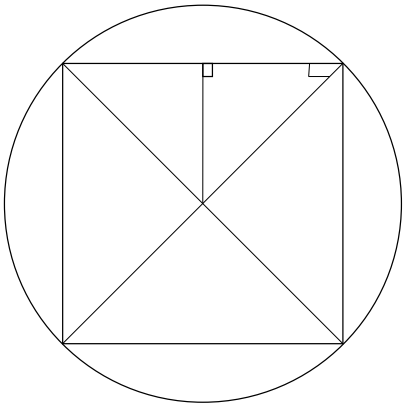
or

$$b=2a$$

The only possible relationship between a and b is

$$a=\frac{b}{2}\ \text{ or }\ b=2a.$$

Qn. 4. Using the figure below



Let r be the radius of the circle then the inscribed square is composed of four isosceles right angles with both sides equal to r. The area of the circle is πr^2 and the area of the square is 4· Area of each triangle thus Area of square $= 4 \cdot (\frac{1}{2}r^2) = 2r^2$.

The ratio required is $\pi r^2 : 2r^2 \Rightarrow \pi : 2$.

Qn. 5. In one minute they can do slash $\frac{1}{40} + \frac{1}{30}$ of the job. This is equivalent to

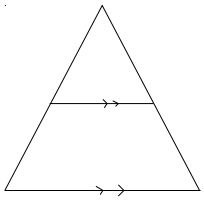
$$\frac{3+4}{120} = \frac{7}{120} \Rightarrow$$

the number of minutes to do the job combined is the reciprocal of this rate i.e $\frac{120}{7}$.

Qn. 6. Average speed $=$ $\frac{\text{total distance}}{\text{total time}}$. Let D be the distance between her uncle's house and her house. Thus

$$\text{Average speed} = \frac{2D}{\frac{D}{40} + \frac{D}{90}} = \frac{2}{\frac{1}{40} + \frac{1}{90}} = \frac{720}{13} = 55.35 km/hr$$

Qn. 7. See figure.



Triangles CDE and CAB are similar thus

$$CE : CD = EB : AD \ \Rightarrow CE : 4 = 2 : 1$$

implying that

$$\frac{CE}{4} = \frac{2}{1} \text{ and } CE = 8cm.$$

Qn. 8. Total amount of non alcohol liquid is $0.9 \times 1 + 0.7 \times 3 = 0.1 + 2.1 = 3$. The total amount of liquid is $1 + 3 = 4 \Rightarrow$ the percentage of non alcohol liquid $= \frac{3}{4} \times 100 = 75\%$

Qn. 9. Square both sides of

$$\log_x a + \log_a x = 3$$

to get

$$(\log_x a)^2 + 2\log_a x \cdot \log_x a + (\log_a x)^2 = 9$$

this implies that

$$(\log_x a)^2 + (\log_a x)^2 = 9 - 2 = 7.$$

Qn. 10. Let x be the number of items that she received then

$$\frac{480,000}{x-2} - \frac{480,000}{x} = 1,000$$

solving gives $x = 32$.

SECTION B : 10 **marks each**

Qn. 11. Since there are only five four groups that can be formed then,

$$125 + 129 + 131 + 137 + 142 = 4 \text{ (sum of ages) } = 4 \times 166.$$

The youngest is $166 - 142 = 24$ years.

Qn. 12. Note that the numbers

$$1, 2, 4, 8, 16, 32, 64, 128, 256$$

are got from the powers of 2 from $0, 1, 2, \ldots$ to 8. Since the powers of 2 from 0 to 8 are being placed in the square so that the sum in each row and column is the same then the common sum of the powers is

$$\frac{1 + 2 + 3 + \ldots + 8}{3} = 12 \Rightarrow$$

thus the common product is $2^{12} = 4096$.

Qn. 13. Let the distance be d meters and their speeds be S_J, S_I, and S_D for John, Isa and David respectively. The following are possible relationships,

$$\frac{S_I}{S_D} = \frac{d}{d-10}; \frac{S_J}{S_D} = \frac{d}{d-28}$$

and

$$\frac{S_J}{S_I} = \frac{d}{d-20}$$

The relation

$$\frac{S_J}{S_I} \cdot \frac{S_I}{S_D} = \frac{S_J}{S_D}$$

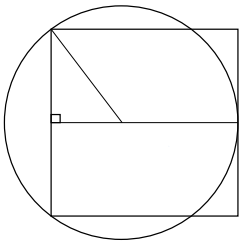
gives

$$\frac{d}{d-20} \cdot \frac{d}{d-10} = \frac{d}{d-28}.$$

On solving we get $d = 100$.

Qn. 14. The units digits of $2^1, 2^2, 2^3$, and 2^4 are respectively $2, 4, 8$ and 6. Continuing the powers of 2 we have $2^5 = 32, 2^6 = 64, 2^7 = 128, 2^8 = 256$. we note therefore that these four digits repeat in the same order all through. The positive integers with increasing powers of 2 so that when the exponent is divisible by 4 units digit is 6. The nearest year to 2017 divisible by 4 is 2016 meaning 2^{2017} is one step into the next cycle whose units digit is 2.

Qn. 15. Given



From the right triangle in the diagram below we have that;

$$r^2 = 1 + (2 - r)^2$$

showing that $r = \frac{5}{4}$ units.